

Idea-IoT-based Solar Powered Pumping and Manuring System for Agriculture
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Problem:

- **Irregular Power Supply and Irrigation:** Inconsistent and lack of power supply and outdated irrigation systems reduce crop yields.
- **Fertilizer and Manure Optimization:** Farmers struggle with effectively using fertilizers and manure, impacting the quality of the harvest.
- **Economic Losses:** These inefficiencies and dependence on other farmers to meet pumping demands lead to significant financial losses for farmers.

Proposed Solution:

- **Solar-Powered Pumping:** Use solar panels to power water pumps directly during daylight, ensuring off-grid irrigation.
- **IoT Monitoring:** Deploy IoT sensors to monitor soil moisture and fertility in real-time, automatically triggering irrigation when moisture is low.

Market Opportunities of proposed business

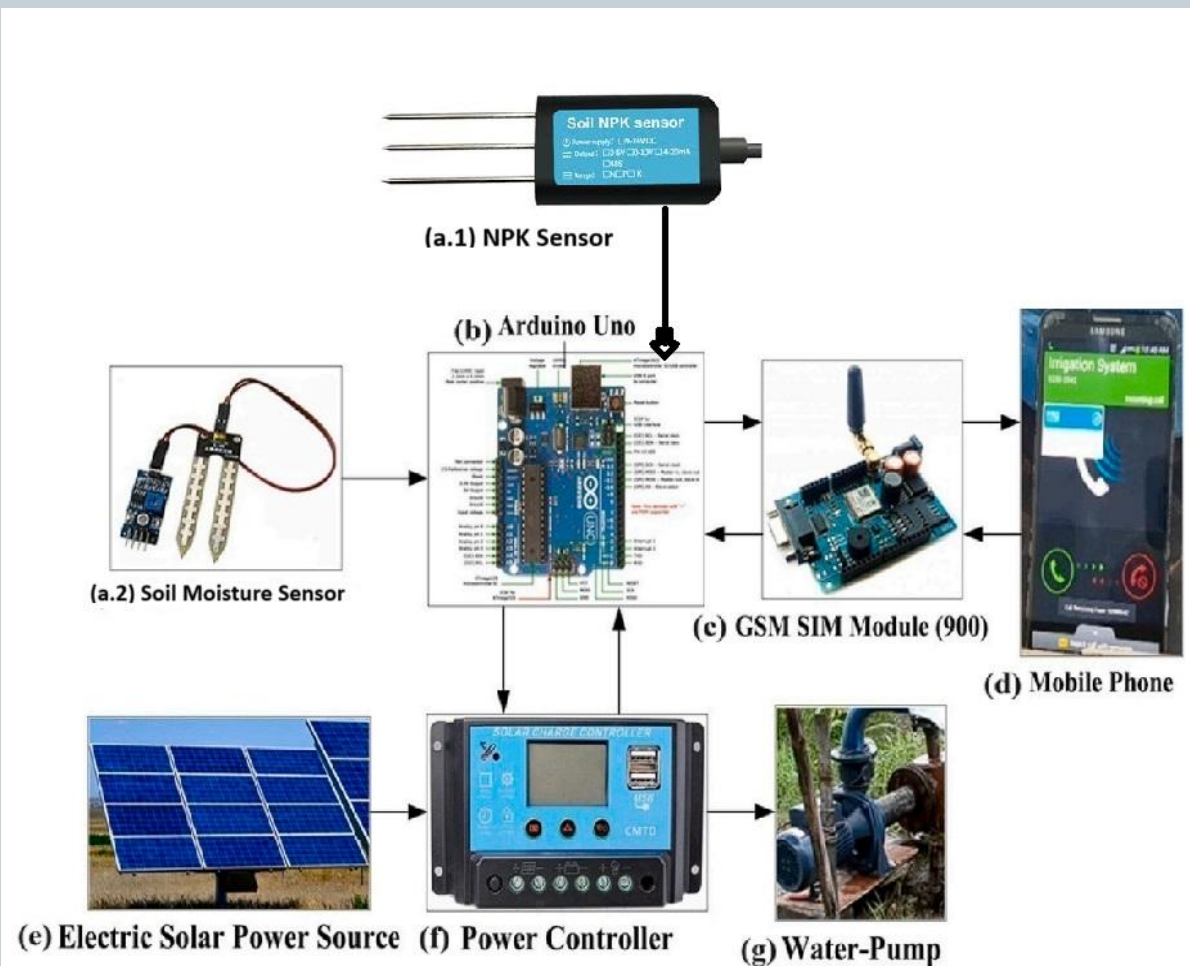


- 893 villages in India are still out of electricity; Whereas more than 20,000 villages still have no regular power supply. Average Indian farmer has a pumping load demand of 1200kwh per month.
- So, on an average at the per unit rate of Rs 5-8.5 the annual running cost of the pumps is Rs 60,000- Rs 80,000.
- Also, farmers are in crucial need of proper irrigation and fertilizing systems to make their produce uniform.
- In existing automated irrigation systems, there is a lack of regular maintenance services and the issue of precision in manuring process.

2 HP	400 Units/Month
3 HP	600 Units/Month
5 HP	1000 Units/Month
15 HP	2000 Units/Month

Fig1: Rating wise pump load demand*

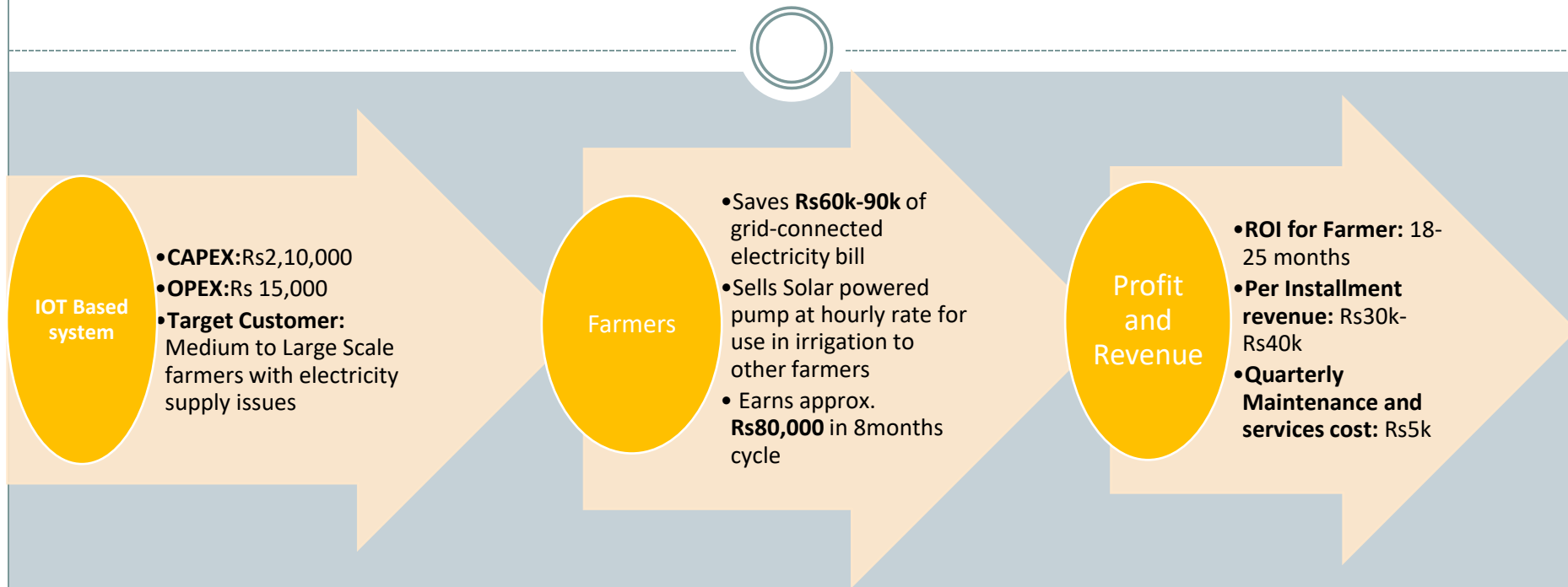
Distinctive Competence of proposed Business



- Existing IOT based systems are lacking the integration of crop specific sensor-based systems.
- Also, the ROI for the farmer is usually >5 years
- But in our system the load demands of other neighboring farms are met by the Solar system in exchange of Rs100/hour

Fig2: Complete Model in a Glance

Value proposition of your business

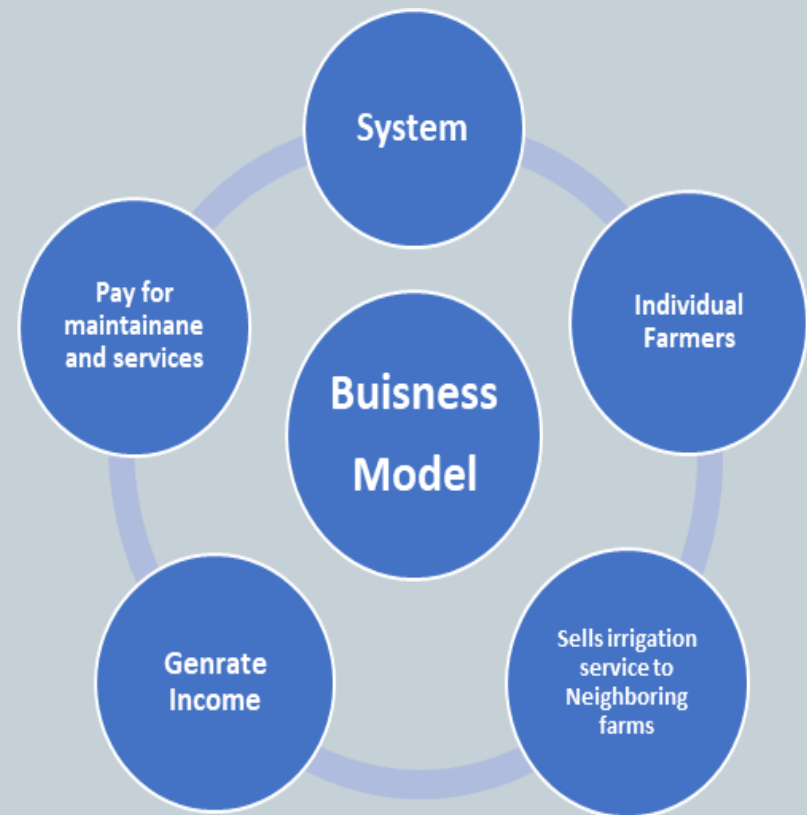


- Farmer will benefit more from this than existing irrigation systems deployed
- They can earn their money back in a period of time.
- Farmers not only save their money by installation of the system but also the yield will increase for sure
- The maintenance services will be provided twice in a produce period which is absent in other service providers

Business Model :-

Type	Value
Prime Consumer	Medium to Large scale farmers
Total payment	Half after the solar setup installation
Mode	Via loans or Direct payment

Type	Value
Cover Revenue	~Rs30,000
Cost	Rs 2,20,000
Gross margin(as % of revenue)	67%
Net Cash Flow	Rs40,000



Risk factors to execution in proposed business



- High initial setup costs may deter adoption.
- Technology failures could disrupt irrigation.
- Maintenance and repair of solar panels and IoT devices require expertise.
- Solar power generation depends on weather conditions.
- Farmers need basic awareness to use IoT systems effectively.
- Competition from existing products
- Economic conditions may affect farmers' ability to pay for shared irrigation.

Requirements of funds and use for proposed business

Financial Requirements:

Sr. no	Components	Expenses
1	Polycrystalline Solar Module Panel, TP330	₹120000(15 units)
2	Soil Moisture Sensor	₹17500(5 units)
3	NPK sensor	₹12000(3 unit)
4	3HP Solar Pump and Controller	₹22000
5	Support Structure	₹15000
6	Service & Miscellaneous	₹33500
7	Total	₹220,000*

Fig3:The upfront capital investment required for each system is ₹220,00.

Revenue Generation & Return on Investment (ROI):

- **Primary Revenue:** Personal use of the system to irrigate the farmer's own crops, saving on water and energy costs.
- **Secondary Revenue:** Offering irrigation services to neighboring farms. This shared-use model enables farmers to earn an additional income, which can be used to offset the initial investment over time.
- Based on current market conditions, the system is expected to generate an additional **₹8000 - ₹10,000 per month** from neighboring farms, with the potential to recover the investment in **18-25 months**.
- **Operational & Maintenance Costs:**
- **Maintenance:** Approximately **₹5,000** per year for upkeep of solar panels, sensors, and pump systems.
- **Contingency:** Allocating **5-10%** of the total cost for potential component replacements (e.g., sensors or pump).

***Inclusive of all GST**